



AmeraKey®

Quantum-Proof Encryption

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Introduction

The advancement of quantum computer technology is rapidly rendering current encryption technologies useless. Great effort and considerable cost are being invested in finding solutions, with the focus concentrated on salvaging, enhancing, or developing new asymmetric encryption algorithms. These algorithms, however, depend upon a mathematical relationship between a public and private key. As such, regardless of how obtuse the relationship, an attack algorithm will eventually be discovered, because math is math.

Ameralot, Inc. has developed a technology called **AmeraKey®**, which provides a user-friendly, frictionless, and highly secure methodology for generating and distributing symmetric encryption keys. **AmeraKey** is a unique, patented technology (14 issued patents) that solves the problem of distributing symmetric keys to remote users and eliminates encryption keys based on insecure passwords.

AmeraKey relies on familiar user activities, such as selecting and sending a picture and entering a passcode PIN (which users already do with banks, ATMs, and other applications). Selecting the picture and passphrase pin is done only once (at registration). Afterward, **AmeraKey** delivers thousands of encryption keys, allows those keys to change on the fly, and eliminates the need for passwords. With applications in enterprise, data collaboration (including data in transit and data at rest), communications (including video conferencing, text messaging, and email), finance, and IoT, **AmeraKey** can provide true frictionless and quantum-proof security.

Quantum-Safe Cryptography

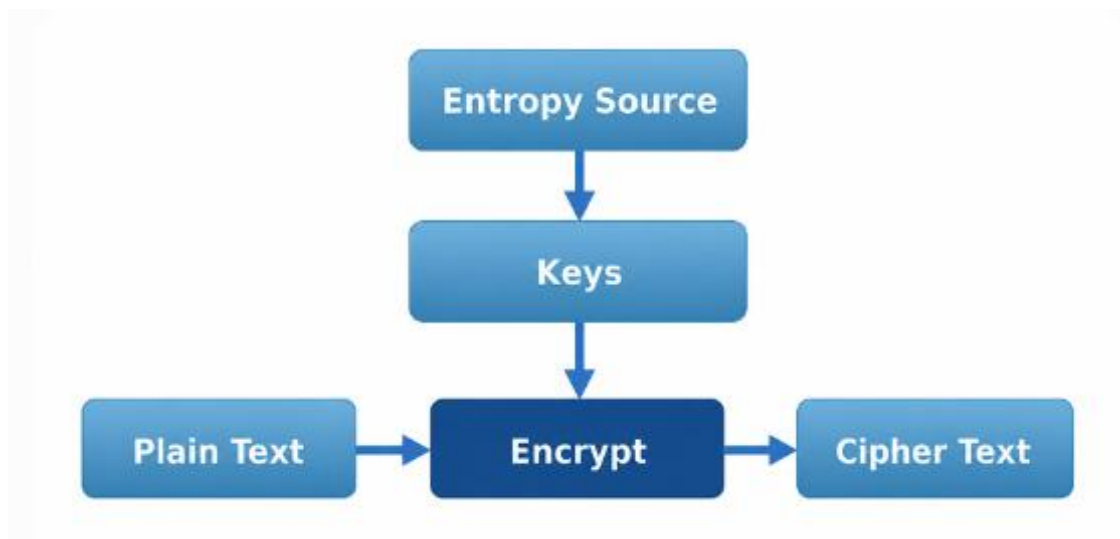


Figure 1 - Basic Encryption

Encryption is the process of scrambling data into a form that is unreadable to humans. There are two basic methodologies for encryption: symmetric encryption and asymmetric encryption. The most secure level of encryption is the one-time pad, which is what symmetric encryption tries to emulate. The security of symmetric encryption lies in the fact that only the sender and receiver possess the encryption keys, which are not exposed to the outside world and cannot be derived. Also, symmetric keys are typically shorter,

thus allowing large files to be encrypted with fewer computing resources. However, the drawback is the distribution of keys between two parties that are remote from each other.

Asymmetric encryption relies on a scheme of public and private keys. The public key is a mathematical derivation of the private key such that a public key can encrypt, while only a private key can decrypt. This solves the issue of key sharing as person A would encrypt a message with person B's public key, and only person B can decrypt it with his private key. The public key is shared with everyone, while the private key is never shared, thereby solving the problem presented by symmetric encryption. However, the penalty for this is that the keys are longer, making the process computationally intensive and not suitable for large files.

Symmetric and asymmetric encryption also differ in their ability to withstand attacks from quantum computers. In an article on asymmetric encryption by EETimes ([Quantum Computing is a Challenge for Cryptography by Maurizio Di Paolo Emilio, 10.02.2020](#)):

"The RSA and ECC algorithms are practically impossible to decipher with brute force methods on a traditional computer. But they're not impossible for a quantum computer with sufficiently stable qubits. Quantum computers can therefore threaten all communications, trade, and finance."

Accordingly, asymmetric encryption is not quantum safe. Any asymmetric encryption scheme that depends on a mathematical relationship between public and private keys is susceptible to a quantum attack. However, symmetric encryption has been deemed to be quantum safe. According to an ETSI (European Telecommunications Standards Institute) white paper on [quantum safe cryptography and security](#) – page 13:

"Symmetric key ciphers like AES are believed to be quantum-safe, whereas many public key ciphers like RSA are known not to be. A protocol that relies exclusively on ciphers like RSA is vulnerable to quantum attack, but a protocol that can adapt to use quantum-safe ciphers is considered quantum-safe."

A One-time pad (OTP) is a special case of symmetric encryption. It's unique in that it provides perfect secrecy when used correctly. In symmetric encryption, the same key is used for encryption and decryption. OTP follows this principle but has stricter requirements:

- The encryption key must be at least as long as the plaintext message.
- The key must be generated using true randomness, not predictable algorithms.
- Each key can be used only once; reusing it compromises security.
- The key must be kept private.
- If all conditions are met, an OTP encryption cannot be broken.

AmeraKey® Quantum-Safe Cryptography

The **AmeraKey** solution for quantum-safe cryptography enables users to share a picture and a PIN simply. From that exchange, each user can independently generate large pools of entropy from which they can extract thousands of high-entropy encryption keys to be used with symmetric algorithms, such as AES, or utilize that entropy pool as part of a one-time pad implementation. The entropy pool and any keys or one-time pads are neither stored nor transmitted between parties, resolving the problem of third-party key storage and rendering man-in-the-middle intercepts useless. Synchronization between parties is

achieved through the sharing of an encrypted token. As part of this synchronization, new entropy pools can be generated on the fly during or after a data session. Users can change pictures and passphrase pins during the data session and generate a brand-new set of entropy pools. Clever use of the encryption token will allow users to change the method of harvesting pixels and the output format of the resulting harvest. Entropy pools can be generated using multiple images, resulting in larger pools with higher entropy. With **AmeraKey®**, the user is always in control of the entropy pool. The pool is never transmitted outside of the local user device, and it is entirely device-independent, as it does not rely on device models. With **AmeraKey®**, the problem of key sharing presented by symmetric encryption, the storage of keys by third parties, and the myriad issues associated with asymmetric encryption are resolved.

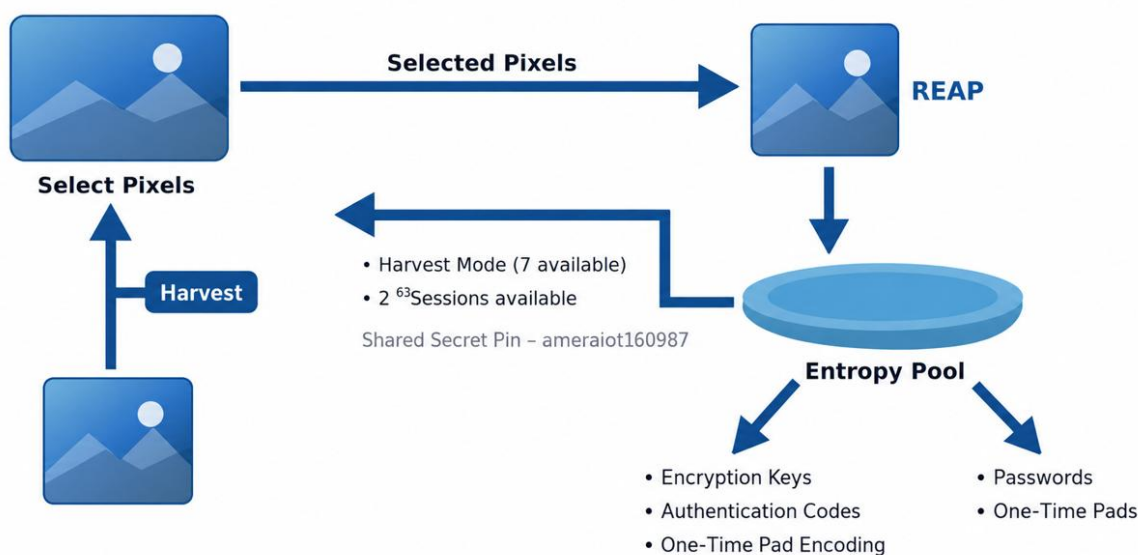


Figure 2 - AmeraKey® Entropy Pool Creation

Entropy Pool Creation

AmeraKey is a source of true randomness that provides the entropy necessary to generate high-quality entropy pools from which encryption keys used by symmetric algorithms such as AES may be extracted or form the basis for various OTP implementations. Like any true random number generator (TRNG), **AmeraKey** extraction takes a snapshot of a natural phenomenon, such as noise in a digital image. To distill the entropy from the noise in an image, the harvesting looks at only low-order bits, with a default of 4 low-order bits from different pixels/colors being exclusive-OR 'ed to produce one bit of key data. This isolates the inherent noise from the subject and the camera from the ordered signal of the photograph, concentrating on the low-level noise. The sampled pixels are randomized by a shuffling process driven by picture data and a hash of the harvest code. This disperses any local trend to similar pixels.

The process involves a two-step approach, where a pixel is selected, and its entropy is extracted. Unlike with asymmetric algorithms, there is no mathematical relationship between the two.

This extraction distills and concentrates the entropy in a way suitable for an RNG. This technique of concentrating entropy is typical for natural sources of entropy, where the raw source data is dispersed.

Any image contains a finite quantity of original noise or entropy. The extraction or distillation process accesses pixels in a sequence based on an internal pixel-driven shuffling algorithm and a hash pattern derived from user input and then manipulates bits from those pixels to concentrate the entropy. The extraction process does not allow pixels to be sampled more than once. The shuffling algorithm may be executed in one of seven harvest modes. Each mode will yield a different entropy pool. Users can select any of the seven harvest modes according to their specific requirements.

In addition to the seven harvest modes, a parameter known as a session variable is utilized in the pixel selection process. Each session variable can generate a single entropy plane (a single harvest), and the session variable is 63 bits long. Thus, the session variable yields 2^{63} (9.2 quintillion) independent entropy planes, which make up the total available entropy pool.

An example of the image harvest is shown in Figures 3 and 4. Running the harvesting algorithm on a typical consumer laptop computer produces output for two different image sizes and harvest modes. These are then compared to a theoretically perfect random number. The size of the picture (measured in pixels) determines the maximum number of keys that can be harvested in a single harvest. As can be seen, the entropy of the generated random keys approaches that of a theoretically random number, with the statistical distribution of the key elements approaching the ideal.

Image / Pool	Pool Size	Keys	Harvest Time	Entropy	Mean	Monte Carlo Pi	Serial Correlation
25 MP image	5.1 Mb	20,003	0.433 sec	7.99974	127.482	3.14002	0.00074087
Perfect Randomness	—	—	—	8	127.5	3.14159	0
6 MP image	1 Mb	3,906	0.059 sec	7.99851	127.496	3.14036	-0.00046759

Figure 3 - Real world example with 25M pixel and 6M pixel picture – Harvest Mode 1

Image / Pool	Pool Size	Keys	Harvest Time	Entropy	Mean	Monte Carlo Pi	Serial Correlation
25 MP image	40.97 Mb	160,031	.434 sec	7.99996	127.531	3.14151	0.00061163
Perfect Randomness	—	—	—	8	127.5	3.14159	0
6 MP image	8 Mb	31,250	0.061 sec	7.99981	127.664	3.13734	-0.0308025

Figure 4 - Real world example with 25M pixel and 6M pixel picture – Harvest Mode 2

As previously stated, the benefit of the **AmeraKey** approach is the ability to generate large entropy pools, which enable the simultaneous generation of thousands of encryption keys of any size at both the sender and the receiver within a single computation. All that is ever shared is the picture and the passphrase PIN. Users can change keys on the fly during or after a data session. They can even change pictures and passphrase pins during the data session and generate a brand-new entropy pool. Users can change both the method of harvesting pixels and the output format of the resulting harvest, allowing the generation of thousands of additional keys from the same picture. Multiple pictures may be harvested together, increasing both the number of keys and entropy. With this technology, **Amera** Picture and PIN provides

a mechanism to synchronize true randomness across multiple remote locations, enabling true quantum-safe cryptography.

AMERA® OTPE

One-time pads represent the ultimate encryption security. When done correctly, they cannot be broken. However, the requirements for generating a proper one-time pad place real-world restrictions on their use:

- The encryption key must be at least as long as the plaintext message –

This tends to restrict the size of the message that can be sent, as long messages or large files would require a huge entropy pool.

- The key must be generated using true randomness, not predictable algorithms –

This restricts how the one-time pad key is generated

- Each key can be used only once; reusing it compromises security.

This necessitates the need for large entropy pools.

Amera has developed and patented a one-time pad encoding scheme that addresses these issues.

The picture and the user-provided passphrase PIN may be used in one-time pads in two ways. First, it is a method for generating a one-time random key to encrypt or decrypt the message. The second method involves utilizing the entries in an entropy pool to encode the message, without relying on mathematical functions such as addition or XOR; instead, it uses the byte location as a means of encoding. The distances between the relevant bytes are nibble encoded and form the basis of the message. The implications of this second approach are astounding, in that not only are encryption keys and entropy pools never exchanged between the sender and receiver, but the message itself is also never sent between them. The only thing sent between two parties is the relative distances between bytes, such that an adversary would need to construct

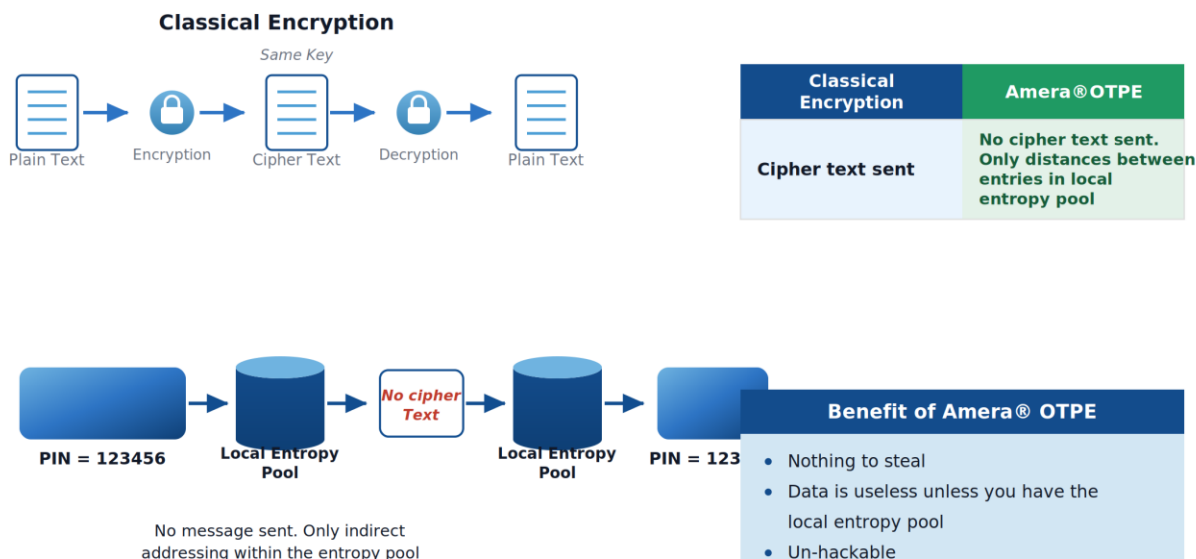


Figure 5 - AMERA OTPE vs Standard Encryption

the entire entropy pool to perform a decode. What makes this possible is the unique way in which the entropy pools are created. Each harvest creates what is termed an entropy plane. There can be a total of 2^{63} (9.2 quintillion) independent entropy planes that make up the total available entropy pool. However, not all of these entropy planes are required for any message. Only enough entropy planes to satisfy the message length constraint need to be generated, and these are generated only as needed, and only one entropy plane is resident in memory at any one time. Thus, we see that conditions 1 and 3 for one-time pads are satisfied. The picture and passphrase pin methodology serves to satisfy the randomness requirement of condition 2.

To further thwart adversaries, the starting session variable can be anything, and the increment of the session variable could also be anything.

Figure 6, taken from [Amera Patent US12225125](#), is a simplified flow chart depicting the **OTPE** process, and Figure 7 (also from [Amera Patent US12225125](#)) illustrates how the encoding and decoding work for a simple text message, and Figure 8 depicts the nibble encoding scheme.

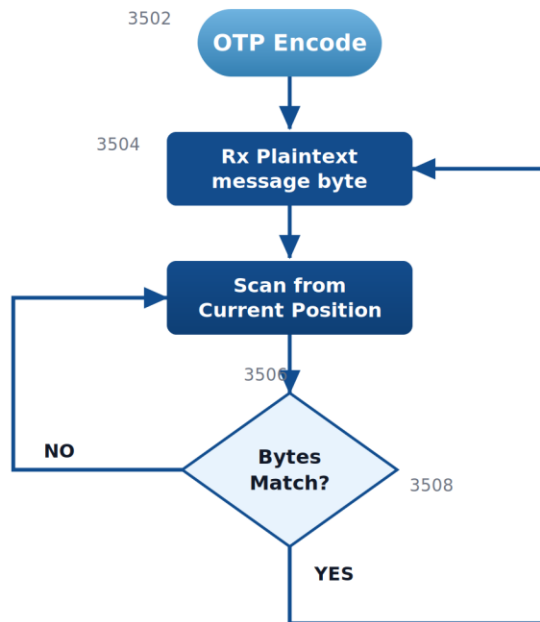


Figure 6 - OTPE basic flow

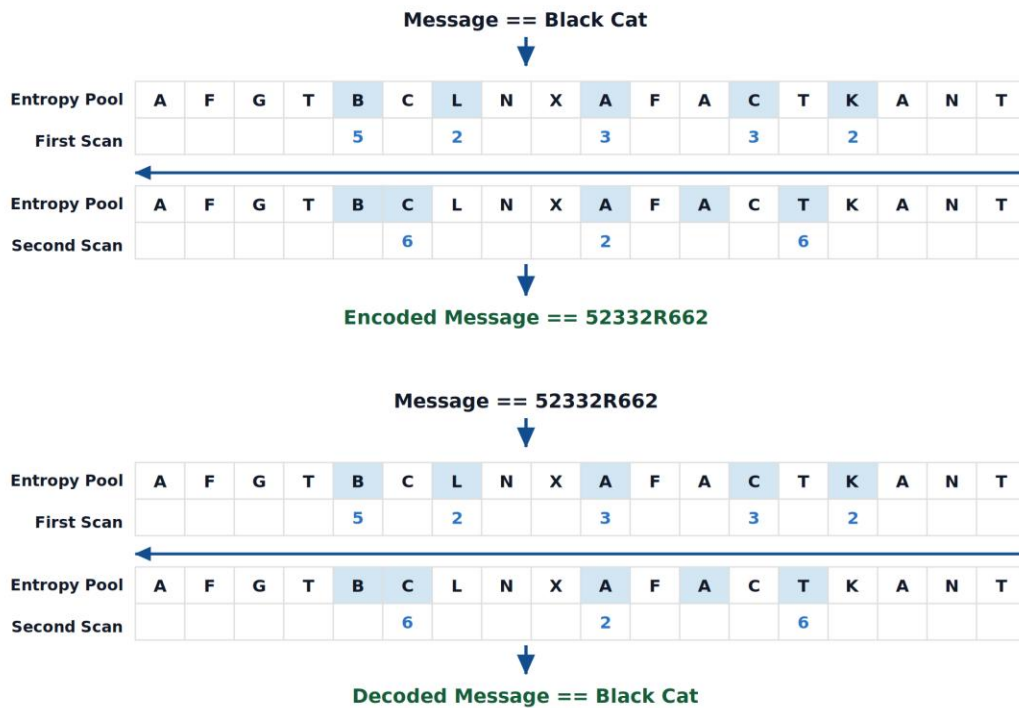


Figure 7 - OTPE Encoding and Decoding

Encoding of the word – Text	T	e	x	t
Position in Entropy Plane	3	9	11	25
Distance	3	6	2	14
Nibble Value	3	6	2	E1 **

Figure 8 - OTPE Nibble Encoding

Key Management



Figure 9 - RSA Key Distribution

The fundamental issue with symmetric encryption is key management. In symmetric encryption, both parties to the communication channel must have the same key, so the problem becomes how party B obtains the key that party A used to encrypt the message. The traditional way is to use a hybrid approach where a symmetric key is generated, then encrypted with RSA and transmitted to party B. However, we know that asymmetric algorithms, such as RSA, are not quantum-proof and can be broken. If an adversary breaks that encryption, they now have access to the symmetric key. With **Amera's** technology, identical entropy pools are created independently on both sides of the communication channel, thus eliminating the need to transmit a symmetric key. Synchronization is accomplished using an encrypted token. The token can be viewed as a recipe that the **Amera** engine uses to generate the appropriate entropy pool. This allows both sides of the communication channel to generate the required entropy pool independently. This technique enables the encryption key to be changed at any time, before, during, or after communication, simply by transmitting a new encrypted token.



Figure 10 - Amera Key Distribution

The question now arises on how one obtains and secures the initial image and PIN. While this approach is implementation-dependent, the safest method is through physical distribution. Physical distribution may take the form of a specific hardware device, such as a USB drive, or a pre-provisioned device or app. As an additional precaution, the initial image and PIN could be set up for one-time use, requiring the user to select a new image and PIN after the first use. Various pictures and PINs would be encrypted and stored in an encrypted database either in a secure memory location or on a physically removable USB drive.

Authentication

A brief note regarding authentication: Since **AmeraKey** generates entropy pools, the same method used to extract encryption keys can also produce secure authentication codes. With the user supplying just a picture and a PIN, and utilizing the random bit harvest described above, **AmeraKey** can generate thousands of unique authentication codes. By using a different image or adjusting the harvest parameters, not only can a user authentication code be generated, but a device authentication code can also be produced, allowing a user to authenticate themselves only with an authenticated device. Similarly to encryption, an authentication token is passed between devices to synchronize which of the thousands of authentication codes to use. Authentication codes can be changed on the fly during or after a data session. This information exchange is encrypted using **AmeraKey®**'s encryption technology.

Authentication can be multi-directional, i.e., User to Host, Host to User, Simultaneous Host User, and similarly with devices. Within each data session, a unique encryption token, User authentication code, and device authentication code can be changed. A Session Token is also assigned to a particular data session. This Session Token must match a similarly generated Session Token on the server to which communication is being established. Authentication is only valid if the User ID, User authentication code, device authentication code, and session ID are matched.

Many authentication schemes still rely on either third-party authentication or password authentication. With **AmeraKey®** Authentication, both third-party authentication and passwords are eliminated. The User opens the App and provides either a biometric or a PIN, which serves the purpose of authenticating the User to the App. This action will automatically send the User ID, User, and device authentication codes, as well as Session Tokens, to the Host where authentication is performed. Depending on the application's requirements, with **AmeraKey**, the User can authenticate the Host, the Host can authenticate the User, or they can authenticate each other simultaneously.

Device to Server System Example

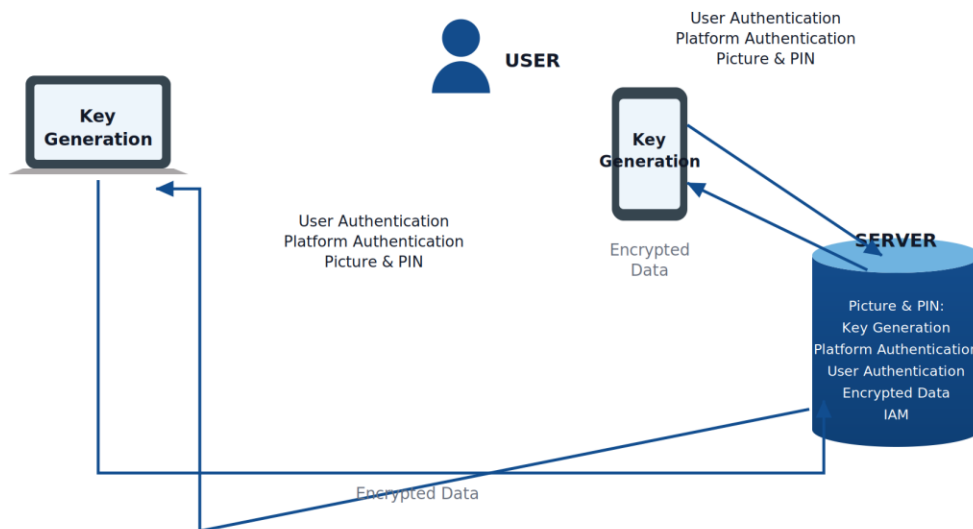


Figure 11 - AmeraKey® Key Generation, Device and User Authentication

A Note about AI

While this paper has been focused on the threat of quantum computers, another looming threat is AI. AI can automate brute force attacks, making them more effective. It can be utilized in chatbots and deepfake technologies to deceive users into disclosing sensitive information. It can allow hackers to scan for weaknesses in networks and software more effectively. It can train malware to avoid detection. It can also analyze large amounts of data to identify patterns, enabling more precise attacks.

Amera's technology can address some of these concerns. **AmeraKey** technology can generate 9.2 quintillion entropy planes, each containing thousands of highly random keys, making brute-force attacks more challenging, even with the aid of AI. The ability to rapidly change encryption keys – even on the fly – adds to the difficulty of AI determining patterns. Proper attention to the quality of the image, particularly in terms of pixel variations, will effectively eliminate any correlation between keys and disrupt an AI attack. With proper use of user and device authentication, phishing attacks can be minimized. All of which serves as a barrier to AI intrusion.

Summary

We have shown how **AmeraKey** technology provides a quantum-hardened solution by:

- Creating large entropy pools with almost perfect entropy
- Utilizing those pools to create secure encryption keys to be used with symmetric encryption algorithms
- Utilizing those pools to enable **Amera's** One-Time Pad Encoding technology

- Never storing or sending encryption keys
- Never sending the message itself utilizing **Amera OTPE**
- Solving the issue of symmetric key management
- Using the entropy pools to create robust authentication

Amera®, through its unique patented **AmeraKey** technology, serves applications in enterprise, data collaboration (including data in transit and data at rest), communications (including video conferencing, text messaging, and email), financial, and IoT, and is the premier solution to providing quantum-proof security.